Twisted representations associated to general automorphisms of a vertex operator algebra

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Abstract

We introduce a notion of strongly $\mathbb{C}^\times$-graded generalized g-twisted V-module for an automorphism g, not necessarily of finite order, of a vertex operator algebra V. For a vertex operator algebra V satisfying $V_{(0)} = \mathbb{C}\mathbf{1}$ and $V_{(n)} = 0$ for $n < 0$ and for $u \in V_{(1)}$ such that the operator $\mathrm{Res}_x Y(u, x)$ has only real eigenvalues and such that the sizes of the Jordan blocks of $\mathrm{Res}_x Y(u, x)$ on $V_{(n)}$ for $n \in \mathbb{Z}$ are bounded, $e^{2\pi i \mathrm{Res}_x Y(u,x)}$ is an automorphism of V. In this case, a strongly $\mathbb{C}^\times$-graded generalized $e^{2\pi i \mathrm{Res}_x Y(u,x)}$-twisted V-module is constructed from a strongly $\mathbb{C}^\times$-graded generalized V-module with a compatible action of g using the exponentiation of the negative power part of the vertex operator $Y(u, x)$. An important feature is that we have to work with generalized (twisted) V-modules which are graded by generalized eigenspaces (not just eigenspaces) for $L(0)$ and the twisted vertex operators in general involve the logarithm of the formal variable.

1 Introduction

Orbifold theories are important examples of conformal field theories obtained from know conformal field theories and their automorphisms (see [DHVW1], [DHVW2], [DFMS], [DVVV], [DGM], as well as, for example, [KS], [FKS], [Ba1], [Ba2], [BHS], [dBHO], [HO], [GHHO], [Ba3] and [HH]). Mathematically, the study of orbifold theories is equivalent to the theory of twisted representations of vertex operator algebras, in the sense that any result or

conjecture on orbifold conformal field theories can be formulated precisely as a result or conjecture in the theory of twisted representations of vertex operator algebras.

Twisted modules for vertex operator algebras arose in the construction of the moonshine module vertex operator algebra $V^\natural$ by I. Frenkel, J. Lepowsky and A. Meurman [FLM1], [FLM2], [FLM3]. In fact, $V^\natural$ was the first example of orbifold theories. Given a vertex operator algebra V and an automorphism g of V of finite order, the notion of g-twisted V-module is a natural generalization of the notion of V-module such that the action of g is incorporated. There have been a number of papers constructing and studying such twisted modules for vertex operator algebras (see in particular, [Le1], [Le2], [FLM3], [D], [DL], [DonLM1], [DonLM2], [Li], [BDM], [DoyLM1], [DoyLM2], [BHL]). However, as far as the author knows, the precise notion of g-twisted V-module for an infinite-order automorphism g has not been previously formulated and such twisted modules have not been previously constructed and studied.

The main difficulty in formulating such a general notion of g-twisted V-module is that the right-hand side of the twisted Jacobi identity and the side of the weak associativity containing the iterate of twisted vertex operator map cannot be generalized to the general case. In the present paper, we first reformulate the notion of g-twisted V-module in the case of finite order g using complex variables and the duality properties and then generalize this equivalent definition to give a notion of strongly $\mathbb{C}^\times$-graded generalized g-twisted V-module for a general automorphism g of V. It is important to note that in general strongly $\mathbb{C}^\times$-graded generalized g-twisted V-modules that we introduced in this paper are direct sums of generalized eigenspaces (not just eigenspaces) for $L(0)$ and the twisted vertex operators in general involve the logarithm of the formal variable.

In general, even when g is a finite-order automorphism of V, how to construct explicitly a g-twisted V-module is still an open problem. The problem of constructing strongly $\mathbb{C}^\times$-graded generalized g-twisted V-module for a general automorphism of V is certainly more difficult. However, when we consider certain special types of automorphisms of V, we might still be able to construct strongly $\mathbb{C}^\times$-graded generalized twisted V-modules associated to these automorphisms even though the general problem is not solved.

Given $u \in V_{(1)}$, it is easy to see that $e^{2\pi i \mathrm{Res}_x Y(u,x)}$ is an automorphism of V and , in general, its order might not be finite. In the present paper, we construct a strongly $\mathbb{C}^\times$-graded generalized $e^{2\pi i \mathrm{Res}_x Y(u,x)}$-twisted V-module

for $u \in V_{(1)}$ from a generalized V-module. Our construction generalizes the construction of twisted V-modules by Li [Li] in the special case that the vertex operator algebra contains a Heisenberg vertex operator subalgebra. We use the exponentiation of the negative power part of the vertex operator $Y(u, x)$. Our construction shows the importance of studying exponentials of vertex operators.

Given an abelian group A, strongly A-graded generalized V-modules and logarithmic intertwining operators were studied by Lepowsky, Zhang and the author in [HLZ1] and [HLZ2]. Many of the general results obtained there can be applied to the study of strongly $\mathbb{C}^\times$-graded generalized g-twisted V-modules introduced and constructed in the present paper.

The present paper is organized as follows: In Section 2, we reformulate the notion of g-twisted V-module in the case of finite-order g using complex variables and duality properties. Then we introduce the notion of strongly $\mathbb{C}^\times$-graded generalized g-twisted V-module in the general case in Section 3. In Section 4, we study the exponentials of negative power parts of vertex operators associated to weight 1 elements. The construction of strongly $\mathbb{C}^\times$-graded generalized $e^{2\pi i \mathrm{Res}_x Y(u,x)}$-twisted V-modules for $u \in V_{(1)}$ is given in Section 5.

2 Twisted modules for finite-order automorphisms of a vertex operator algebra

The notion of g-twisted module associated to a finite order automorphism g of V is formulated using the properties of twisted vertex operators obtained in [Le1], [FLM2] and [Le2]; in particular, it uses a twisted Jacobi identity (see (2.1) below) as the main axiom. For an infinite-order automorphism g of V, we cannot write down a generalization of the twisted Jacobi identity in the case of finite-order automorphisms. The twisted Jacobi identity can also be replaced by the weak commutativity and (twisted) weak associativity in the case of finite-order automorphisms (see (2.3) below). But the side of the weak associativity containing the iterate of the twisted vertex operator map cannot be generalized to the case of an (necessarily infinite-order) automorphism which acts on the vertex operator algebra or the twisted module to be defined nonsemisimply. These difficulties force us to use the complex variable approach. To motivate our notion of g-twisted module for a general

automorphism g of V, we reformulate the definition of g-twisted module in terms of complex variables in this section.

Let $(V, Y, \mathbf{1}, \omega)$ be a vertex operator algebra and g an automorphism of V of order $k \in \mathbb{Z}_+$. Then $V = \coprod_{j\in\mathbb{Z}/k\mathbb{Z}} V^j$ where $V^j = \{v \in V \mid gv = \eta^j v\}$ for $j \in \mathbb{Z}/k\mathbb{Z}$, where $\eta = e^{\frac{2\pi\sqrt{-1}}{k}}$. We first recall the notion of g-twisted V-module.

Definition 2.1. A *g-twisted V-module* W is a $\mathbb{C}$-graded vector space $W = \coprod_{n\in\mathbb{C}} W_{(n)}$ equipped with a linear map

$$\begin{aligned} Y^g : V \otimes W &\to W[[x^{1/k}, x^{-1/k}]] \\ v \otimes w &\mapsto Y^g(v, x)w = \sum_{n\in\frac{1}{k}\mathbb{Z}} Y^g_n(v)wx^{-n-1} \end{aligned}$$

satisfying the following conditions:

1. The *grading-restriction condition*: For each $n \in \mathbb{C}$, $\dim W_{(n)} < \infty$ and $W_{(n+l/k)} = 0$ for all sufficiently negative integers l.

2. The *formal monodromy condition*: For $j \in \mathbb{Z}/k\mathbb{Z}$ and $v \in V^j$, $Y^g(v, x) = \sum_{n\in j/k+\mathbb{Z}} Y^g_n(v)x^{-n-1}$.

3. The *lower-truncation condition*: For $v \in V$ and $w \in W$, $Y^g_n(v)w = 0$ for n sufficiently large.

4. The *identity condition*: $Y^g(\mathbf{1}, x) = 1$.

5. The *twisted Jacobi identity*: For $u, v \in V$ and $w \in W$,

$$\begin{aligned} &x_0^{-1}\delta\left(\frac{x_1 - x_2}{x_0}\right) Y^g(u, x_1)Y^g(v, x_2) \\ &\qquad -x_0^{-1}\delta\left(\frac{x_2 - x_1}{-x_0}\right) Y^g(v, x_2)Y^g(u, x_1) \\ &= \frac{1}{k}\sum_{j\in\mathbb{Z}/k\mathbb{Z}} x_2^{-1}\delta\left(\eta^j\frac{(x_1 - x_0)^{1/k}}{x_2^{1/k}}\right) Y^g(Y(g^j u, x_0)v, x_2). \quad (2.1) \end{aligned}$$

6. The *$L^g(0)$-grading condition*: For $n \in \mathbb{Z}$, let $L^g(n) = Y^g_{n+1}(\omega)$, that, is, $Y^g(\omega, x) = \sum_{n\in\mathbb{Z}} L^g(n)x^{-n-2}$. Then $L^g(0)w = mw$ for $w \in W_{(m)}$.

7. The $L^g(-1)$-*derivative condition*: For $v \in V$,

$$\frac{d}{dx}Y^g(v,x) = Y^g(L(-1)v,x).$$

We denote a g-twisted V-module defined above by (W, Y^g) (or briefly, by W).

This definition of g-twisted V-module is a natural generalization of the definition of V-module given in [FLM3] and [FHL]. The main axiom is the twisted Jacobi identity. The follwoing result is proved by Li [Li]:

Proposition 2.2. *The twisted Jacobi identity in Definition 2.1 can be replaced by the following two properties:*

1. *The weak commutativity or locality: For $u, v \in V$, there exists $N \in \mathbb{Z}_+$ such that*

$$(x_1 - x_2)^N Y^g(u,x_1)Y^g(v,x_2) = (x_2 - x_1)^N Y^g(v,x_2)Y^g(u,x_1). \quad (2.2)$$

2. *The weak associativity: For $u \in V^j$ and $w \in W$, there exists $M \in \mathbb{Z}_+$ such that*

$$(x_2+x_0)^{\tilde{j}/k+M}Y^g(Y(u,x_0v,x_2) = (x_0+x_2)^{\tilde{j}/k+M}Y^g(u,x_0+x_2)Y^g(v,x_2) \quad (2.3)$$

for $v \in V$, where $\tilde{j} \in j$ satisfies $0 \le \tilde{j} < k$.

One advantage of the definition, the weak commutativity and the weak associativity above is that they use only formal variables so that one can discuss the multivaluedness of the correlation functions algebraically. But as we mentioned above this formulation of g-twisted V-module for g of finite order cannot be generalized to the case that g is of infinite order. So we first reformulate this definition using complex variables and a duality property.

We need a complex variable formulation of vertex operator maps and some notations first. For any $z \in \mathbb{C}^\times$, we shall use $\log z$ to denote $\log|z| + \sqrt{-1}\arg z$ where $0 \le \arg z < 2\pi$. In general, we shall use $l_k(z)$ to denote $\log z + 2\sqrt{-1}\pi$ for $k \in \mathbb{Z}$. Let W_1, W_2 and W_3 be $\mathbb{C}$-graded vector spaces. The $\mathbb{C}$-gradings on W_1 and W_2 induce a $\mathbb{C}$-grading on $W_1 \otimes W_2$. For a formal variable $x^{1/k}$, we define its *degree* to be $-1/k$. Then the grading on W_3 and the degree of x together give $W_3[[x^{1/k}, x^{-1/k}]]$ a $\mathbb{C}$-graded vector space structure. Let

$$\begin{aligned} X : W_1 \otimes W_2 &\to W_3[[x^{1/k}, x^{-1/k}]] \\ w_1 \otimes w_2 &\mapsto X(w_1,x)w_2 = \sum_{n \in \frac{1}{k}\mathbb{Z}} X_n(w_1)w_2x^{-n-1} \end{aligned}$$

is a linear map preserving the gradings. For any $w_1 \in W_1$ and $w_2 \in W_2$, we define $X^p(w_1, z)w_2$ for $p \in \mathbb{Z}/k\mathbb{Z}$ to be the elements $X(w_1, x)w_2\Big|_{x^n = e^{nl_{\tilde{p}}(z)}}$ of $\overline{W}_3 = \prod_{n\in\mathbb{C}}(W_3)_{(n)}$ where for $p \in \mathbb{Z}/k\mathbb{Z}$, $\tilde{p}$ is the integer in p satisfying $0 \le \tilde{p} < k$. When $p = 0$, for simplicity, we shall denote $X^0(w_1, z)w_2$ simply as $X(w_1, z)w_2$. For $z \in \mathbb{C}^\times$, we have the linear maps

$$\begin{aligned} X^p(\cdot, z)\cdot : W_1 \otimes W_2 &\to \overline{W}_3, \\ w_1 \otimes w_2 &\mapsto X^p(w_1, z)w_2 \end{aligned} \tag{2.4}$$

and we shall denote $X^0(\cdot, z)\cdot$ simply as $X(\cdot, z)\cdot$. For any $w_3' \in W_3'$,

$$\langle w_3', X^p(w_1, z)w_2\rangle$$

are branches of a multivalued analytic function defined on $\mathbb{C}^\times$. We can view this multivalued analytic function on $\mathbb{C}^\times$ as a single-valued analytic function defined on a k-fold covering space of $\mathbb{C}^\times$. For $p \in \mathbb{Z}/k\mathbb{Z}$, we have a map

$$\begin{aligned} X^p : \mathbb{C}^\times &\to \mathrm{Hom}(W_1 \otimes W_2, \overline{W}_3) \\ z &\mapsto X^p(\cdot, z)\cdot . \end{aligned}$$

We shall call X^p the *p-th analytic branch* of the map X.

In particular, for the twisted vertex operator map Y^g, we have its p-th analytic branch $Y^{g;p}$. In terms of these analytic branches, we have:

Proposition 2.3. *The formal monodromy condition in Definition 2.1 can be replaced by the following condition: For $p \in \mathbb{Z}/k\mathbb{Z}$, $z \in \mathbb{C}^\times$ and $v \in V$,*

$$Y^{g;p+1}(gv, z) = Y^{g;p}(v, z). \tag{2.5}$$

Proof. Let W be a g-twisted V-module satisfying Definition 2.1. Then for $v \in V^j$, we have $gv = \eta^j v \in V^j$. Then by the formal monodromy condition, for $p \in \mathbb{Z}/k\mathbb{Z}$, $z \in \mathbb{C}^\times$ and $v \in V^j$,

$$\begin{aligned} Y^{g;p+1}(gv, z) &= \sum_{n\in\frac{1}{k}\mathbb{Z}} Y_n^g(gv)e^{(-n-1)l_{p+1}(z)} \\ &= \sum_{n\in\frac{j}{k}+\mathbb{Z}} Y_n^g(v)\eta^j e^{(-n-1)l_{p+1}(z)} \\ &= \sum_{n\in\frac{j}{k}+\mathbb{Z}} Y_n^g(v)e^{\frac{2\pi j\sqrt{-1}}{k}} e^{(-n-1)l_{p+1}(z)} \end{aligned}$$

$$
\begin{aligned}
&= \sum_{n\in\frac{j}{k}+\mathbb{Z}} Y_n^g(v)e^{(-n-1)(l_{p+1}(z)-2\pi\sqrt{-1})} \\
&= \sum_{n\in\frac{j}{k}+\mathbb{Z}} Y_n^g(v)e^{(-n-1)l_p(z)} \\
&= Y^{g;p}(v,z).
\end{aligned}
$$

Conversely, assume that W satisfies the modification of Definition 2.1 by replacing the formal monodromy condition with the equivariance condition. Then for $j \in \mathbb{Z}/k\mathbb{Z}$ and $v \in V^j$,

$$
\begin{aligned}
\sum_{n\in\frac{1}{k}\mathbb{Z}} Y_n^g(v)\eta^j e^{(-n-1)l_{p+1}(z)} &= \sum_{n\in\frac{1}{k}\mathbb{Z}} Y_n^g(gv)e^{(-n-1)l_{p+1}(z)} \\
&= Y^{g;p+1}(gv,z) \\
&= Y^{g;p}(v,z) \\
&= \sum_{n\in\frac{1}{k}\mathbb{Z}} Y_n^g(v)e^{(-n-1)l_p(z)} \\
&= \sum_{n\in\frac{1}{k}\mathbb{Z}} Y_n^g(v)e^{\frac{2\pi(n+1)\sqrt{-1}}{k}}e^{(-n-1)l_{p+1}(z)}.
\end{aligned}
$$

So we have

$$
\sum_{n\in\frac{1}{k}\mathbb{Z}} Y_n^g(v)(\eta^j - e^{\frac{2\pi(-n-1)\sqrt{-1}}{k}})e^{(n+1)l_p(z)} = 0,
$$

which implies that

$$
Y_n^g(v)(\eta^j - e^{\frac{2\pi(n+1)\sqrt{-1}}{k}}) = 0
$$

for $n \in \frac{1}{k}\mathbb{Z}$. So we have either $Y_n^g(v) = 0$ or $\eta^j = e^{\frac{2\pi(n+1)\sqrt{-1}}{k}}$, that is, either $Y_n^g(v) = 0$ or $n \in j/k + \mathbb{Z}$. So we obtain $Y^g(v,x) = \sum_{n\in j/k+\mathbb{Z}} Y_n^g(v)x^{-n-1}$. ∎

We shall call (2.5) in Proposition 2.3 the *equivariance property.*

For $n \in \mathbb{C}$, let $\pi_n : W \to W_{(n)}$ be the projection. We have:

Theorem 2.4. *The twisted Jacobi identity in Definition 2.1 can be replaced by the following property: For any $u, v \in V$, $w \in W$ and $w' \in W'$, there exists a multivalued analytic function of the form*

$$
f(z_1,z_2) = \sum_{r,s=N_1}^{N_2} a_{rs} z_1^{r/k} z_2^{s/k}(z_1-z_2)^{-N}
$$

for $N_1, N_2, N \in \mathbb{Z}_+$, such that the series

$$\langle w', Y^{g;p}(u, z_1)Y^{g;p}(v, z_2)w\rangle = \sum_{n\in\mathbb{C}} \langle w', Y^{g;p}(u, z_1)\pi_n Y^{g;p}(v, z_2)w\rangle, \quad (2.6)$$

$$\langle w', Y^{g;p}(v, z_2)Y^{g;p}(u, z_1)w\rangle = \sum_{n\in\mathbb{C}} \langle w', Y^{g;p}(v, z_2)\pi_n Y^{g;p}(u, z_1)w\rangle, \quad (2.7)$$

$$\langle w', Y^{g;p}(Y(u, z_1 - z_2)v, z_2)w\rangle = \sum_{n\in\mathbb{C}} \langle w', Y^{g;p}(\pi_n Y(u, z_1 - z_2)v, z_2)w\rangle \quad (2.8)$$

are absolutely convergent in the regions $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$, $|z_2| > |z_1 - z_2| > 0$, respectively, to the branch

$$\sum_{r,s=N_1}^{N_2} a_{rs} e^{(r/k)l_p(z_1)} e^{(s/k)l_p(z_2)} (z_1 - z_2)^{-N} \quad (2.9)$$

of $f(z_1, z_2)$.

Proof. Assume that W is a g-twisted V-module satisfying Definition 2.1. By Proposition 2.2, there exists $N \in \mathbb{Z}_+$ such that (2.2) holds.

For $w \in W$ and $w' \in W'$, we know that

$$\langle w', Y^g(u, x_1)Y^g(v, x_2)w\rangle \quad (2.10)$$

involves finitely many positive powers of $x_1^{1/k}$ and fnitely many negative powers of $x_2^{1/k}$ and

$$\langle w', Y^g(v, x_2)Y^g(u, x_1)w\rangle \quad (2.11)$$

involves finitely many negative powers of $x_1^{1/k}$ and fnitely many positive powers of $x_2^{1/k}$. Since $(x_1 - x_2)^N$ is a polynomial in $x_1^{1/k}$ and $x_2^{1/k}$, the same statements are also true for

$$(x_1 - x_2)^N \langle w', Y^g(u, x_1)Y^g(v, x_2)w\rangle$$

and

$$(x_1 - x_2)^N \langle w', Y^g(v, x_2)Y^g(u, x_1)w\rangle,$$

respectively. By (2.2), they are acrually equal. Thus they are both equal to

$$\sum_{r,s=N_1}^{N_2} a_{rs} x_1^{r/k} x_2^{s/k} \in \mathbb{C}[, x_1^{1/k}, x_1^{-1/k}, x_2^{1/k}, x_2^{-1/k}].$$

Since (2.10) involves finitely many positive powers of $x_1^{1/k}$ and fnitely many negative powers of $x_2^{1/k}$, it must be equal to the expansion of

$$\sum_{r,s=N_1}^{N_2} a_{rs} x_1^{r/k} x_2^{s/k} (x_1 - x_2)^{-N}$$

as a series of the same form. Thus the expansion of (2.9) in the region $|z_1| > |z_2| > 0$ is equal to (2.6). The same argument shows that the expansion of (2.9) in the region $|z_2| > |z_1| > 0$ is equal to (2.7).

For $j \in \mathbb{Z}/k\mathbb{Z}$, let $\tilde{j} \in j$ satisfying $0 \le \tilde{j} < k$. For $u \in V^j$, by Proposition 2.2 there exists $M \in \mathbb{Z}_+$ such that (2.3) holds.

For $w \in W$ and $w' \in W'$, we know that (2.10) involves finitely many positive powers of $x_1^{1/k}$ and fnitely many negative powers of $x_2^{1/k}$. So

$$\langle w', Y^g(u, x_0 + x_2) Y^g(v, x_2) w \rangle \tag{2.12}$$

involves finitely many positive powers of $x_0^{1/k}$ and fnitely many negative powers of $x_2^{1/k}$. We also know that

$$\langle w', Y^g(Y(u, x_0)v, x_2) w \rangle \tag{2.13}$$

involves finitely many negative powers of $x_0^{1/k}$ and fnitely many positive powers of $x_2^{1/k}$. Since $(x_0 + x_2)^{\tilde{j}/k+M}$ involves finitely many positive powers of $x_0^{1/k}$ and no negative powers of $x_2^{1/k}$ and $(x_2 + x_0)^{\tilde{j}/k+M}$ involves finitely many positive powers of $x_2^{1/k}$ and no negative powers of $x_2^{1/k}$, the same statements above for (2.12) and (2.13) are also true for

$$(x_0 + x_2)^{\tilde{j}/k+M} \langle w', Y^g(u, x_0 + x_2) Y^g(v, x_2) w \rangle$$

and

$$(x_2 + x_0)^{\tilde{j}/k+M} \langle w', Y^g(Y(u, x_0)v, x_2) w \rangle,$$

respectively. Since we have proved that (2.10) is equal to

$$\sum_{r,s=N_1}^{N_2} a_{rs} x_1^{r/k} x_2^{s/k} (x_1 - x_2)^{-N},$$

we obtain

$$\begin{aligned}&(x_0+x_2)^{\tilde{j}/k+M}\langle w', Y^g(u, x_0+x_2)Y^g(v, x_2)w\rangle\\ &\quad=\sum_{r,s=N_1}^{N_2} a_{rs}(x_0+x_2)^{r/k+\tilde{j}/k+M}x_2^{s/k}x_0^{-N}.\end{aligned}$$

By (2.3), we also have

$$\begin{aligned}&(x_2+x_0)^{\tilde{j}/k+M}\langle w', Y^g(Y(u, x_0)v, x_2)w\rangle\\ &\quad=\sum_{r,s=N_1}^{N_2} a_{rs}(x_2+x_0)^{r/k+\tilde{j}/k+M}x_2^{s/k}x_0^{-N}.\end{aligned}$$

Thus the expansion of (2.9) in the region $|z_2| > |z_1 - z_2| > 0$ is equal to (2.8).

Conversely, assume that the property in the theorem holds. Let

$$f_j(x_1, x_2, x_0) = \sum_{r,s=N_1}^{N_2} a_{rs}\eta^{-jr}x_1^{r/k}x_2^{s/k}x_0^{-N}$$

for $j \in \mathbb{Z}/kZ$. Then this property gives

$$\langle w', Y^g(u, x_1)Y^g(v, x_2)w\rangle = f_0(x_1, x_2, x_1 - x_2), \tag{2.14}$$

$$\langle w', Y^g(v, x_2)Y^g(u, x_1)w\rangle = f_0(x_1, x_2, -x_2 + x_1), \tag{2.15}$$

$$\langle w', Y^g(Y(u, z_1 - z_2)v, z_2)w\rangle = f_0(x_2 + x_0, x_2, x_0) \tag{2.16}$$

for $u, v \in V$, $w \in W$ and $w' \in W'$. From (2.14), (2.16) and the equivariance property, we have

$$\langle w', Y^g(Y(g^j u, z_1 - z_2)v, z_2)w\rangle = f_j(x_2 + x_0, x_2, x_0) \tag{2.17}$$

for $u, v \in V$, $w \in W$ and $w' \in W'$.

We have the formal variable identity

$$x_0^{-1}\left(\frac{x_1 - x_2}{x_0}\right) - x_0^{-1}\left(\frac{x_2 - x_1}{-x_0}\right) = \frac{1}{k}\sum_{j\in\mathbb{Z}/k\mathbb{Z}} x_2^{-1}\left(\eta^j\frac{(x_1 - x_0)^{1/k}}{x_2^{1/k}}\right).$$

Using this identity and the property of the delta-function, we have

$$x_0^{-1}\left(\frac{x_1 - x_2}{x_0}\right) f_0(x_1, x_2, x_1 - x_2) - x_0^{-1}\left(\frac{x_2 - x_1}{-x_0}\right) f_0(x_1, x_2, -x_2 + x_1)$$

$$
\begin{aligned}
&= x_0^{-1}\left(\frac{x_1-x_2}{x_0}\right) f_0(x_1,x_2,x_0) - x_0^{-1}\left(\frac{x_2-x_1}{-x_0}\right) f_0(x_1,x_2,x_0)\\
&= \frac{1}{k}\sum_{j\in\mathbb{Z}/k\mathbb{Z}} x_2^{-1}\left(\eta^j\frac{(x_1-x_0)^{1/k}}{x_2^{1/k}}\right) f_0(x_1,x_2,x_0)\\
&= \frac{1}{k}\sum_{j\in\mathbb{Z}/k\mathbb{Z}} x_2^{-1}\left(\eta^j\frac{(x_1-x_0)^{1/k}}{x_2^{1/k}}\right) \sum_{r,s=N_1}^{N_2} a_{rs}x_1^{r/k}x_2^{s/k}x_0^{-N}\\
&= \frac{1}{k}\sum_{j\in\mathbb{Z}/k\mathbb{Z}} x_2^{-1}\left(\eta^j\frac{(x_1-x_0)^{1/k}}{x_2^{1/k}}\right) \sum_{r,s=N_1}^{N_2} a_{rs}((x_1-x_0)+x_0)^{r/k}x_2^{s/k}x_0^{-N}\\
&= \frac{1}{k}\sum_{j\in\mathbb{Z}/k\mathbb{Z}} x_2^{-1}\left(\eta^j\frac{(x_1-x_0)^{1/k}}{x_2^{1/k}}\right)\cdot\\
&\qquad\qquad \cdot\sum_{r,s=N_1}^{N_2} a_{rs}(x_1-x_0)^{r/k}\left(1+\frac{x_0}{x_1-x_0}\right)^{r/k} x_2^{s/k}x_0^{-N}\\
&= \frac{1}{k}\sum_{j\in\mathbb{Z}/k\mathbb{Z}} x_2^{-1}\left(\eta^j\frac{(x_1-x_0)^{1/k}}{x_2^{1/k}}\right) \sum_{r,s=N_1}^{N_2} a_{rs}\eta^{-jr}x_2^{r/k}\left(1+\frac{x_0}{x_2}\right)^{r/k} x_2^{s/k}x_0^{-N}\\
&= \frac{1}{k}\sum_{j\in\mathbb{Z}/k\mathbb{Z}} x_2^{-1}\left(\eta^j\frac{(x_1-x_0)^{1/k}}{x_2^{1/k}}\right) \sum_{r,s=N_1}^{N_2} a_{rs}\eta^{-jr}(x_2+x_0)^{r/k}x_2^{s/k}x_0^{-N}\\
&= \frac{1}{k}\sum_{j\in\mathbb{Z}/k\mathbb{Z}} x_2^{-1}\left(\eta^j\frac{(x_1-x_0)^{1/k}}{x_2^{1/k}}\right) f_j(x_2+x_0,x_2,x_0). \qquad (2.18)
\end{aligned}
$$

From (2.14)–(2.16) and (2.18), we obtain

$$
\begin{aligned}
&x_0^{-1}\delta\left(\frac{x_1-x_2}{x_0}\right)\langle w', Y^g(u,x_1)Y^g(v,x_2)w\rangle\\
&\qquad -x_0^{-1}\delta\left(\frac{x_2-x_1}{-x_0}\right)\langle w', Y^g(v,x_2)Y^g(u,x_1)w\rangle\\
&= \frac{1}{k}\sum_{j\in\mathbb{Z}/k\mathbb{Z}} x_2^{-1}\delta\left(\eta^j\frac{(x_1-x_0)^{1/k}}{x_2^{1/k}}\right)\langle w', Y^g(Y(g^ju,x_0)v,x_2)w\rangle
\end{aligned}
$$

for $u, v \in V$, $w \in W$ and $w' \in W'$. Since $w \in W$ and $w' \in W'$ are arbitrary, we obtain the twisted Jacobi identity (2.1) for $u, v \in V$. ∎

We shall call the property in Theorem 2.4 the *duality property*.

3 Definition of strongly $\mathbb{C}^\times$-graded generalized g-twisted module

We give the definition of strongly $\mathbb{C}^\times$-graded generalized g-twisted V-modules for an automorphism g of V of possibly infinite order in this section. In the case that g is of finite order, g always acts on V semisimply so that we have $V = \coprod_{j\in\mathbb{Z}/k\mathbb{Z}} V^j$. When the order of g is infinite, it does not have to act on V semisimply. In particular, we have to allow not only powers of x in the twisted vertex operators, but also integral powers of $\log x$ in these operators. As we have mentioned in the introduction, the left-hand side of the twisted Jacobi identity cannot be generalized to the case that the order of g is infinite. Also we cannot straightforwardly generalize the formal monodromy condition to the general case. On the other hand, the equivariance property and the duality property in Proposition 2.3 and Theorem 2.4, respectively, can be generalized easily. Our definition will use the generalizations of these axioms.

First we need to further generalize our consideration of analytic branches of the map X in the preceding section. For formal variables x and $\log x$, we define their *degrees* to be -1 and 0. Let W_1, W_2 and W_3 be $\mathbb{C}$-graded vector spaces. Then the grading on W_3 and the degrees of x and $\log x$ together give $W_3\{x\}[\log x]$ a $\mathbb{C}$-grading. Let

$$\begin{aligned} X : W_1 \otimes W_2 &\to W_3\{x\}[\log x] \\ w_1 \otimes w_2 &\mapsto X(w_1, x)w_2 = \sum_{n\in\mathbb{C}} \sum_{k=1}^{K} X_{n,k}(w_1)w_2 x^{-n-1}(\log x)^k \end{aligned}$$

is a linear map preserving the gradings. For any $w_1 \in W_1$ and $w_2 \in W_2$, we define $X^p(w_1, z)w_2$ for $p \in \mathbb{Z}$ to be the elements $X(w_1, x)w_2\Big|_{x^n = e^{nl_{\hat{p}}(z)},\ \log x = l_p(z)}$ of $\overline{W}_3$. When $p = 0$, we shall denote $X^0(w_1, z)w_2$ simply as $X(w_1, z)w_2$. For $z \in \mathbb{C}^\times$, we have the linear maps

$$\begin{aligned} X^p(\cdot, z)\cdot : W_1 \otimes W_2 &\to \overline{W}_3, \\ w_1 \otimes w_2 &\mapsto X^p(w_1, z)w_2 \end{aligned} \tag{3.1}$$

and we shall denote $X^0(\cdot, z)\cdot$ simply as $X(\cdot, z)\cdot$. For any $w_3' \in W_3'$,

$$\langle w_3', X^p(w_1, z)w_2\rangle$$

are branches of a multivalued analytic function defined on $\mathbb{C}^\times$. We can view this multivalued analytic function on $\mathbb{C}^\times$ as a single-valued analytic function defined on a covering space of $\mathbb{C}^\times$. For $p \in \mathbb{Z}$, we have a map

$$\begin{aligned} X^p : \mathbb{C}^\times &\to \mathrm{Hom}(W_1 \otimes W_2, \overline{W}_3) \\ z &\mapsto X^p(\cdot, z)\cdot. \end{aligned}$$

We shall call X^p the *p-th analytic branch* of the map X.

We also need a $\mathbb{C}^\times$-grading structure on V given by an automorphism of V. Since g preserve $V_{(n)}$ for $n \in \mathbb{Z}$ and $V_{(n)}$ for $n \in \mathbb{Z}$ are all finite dimensional, we have

$$V_{(n)} = \coprod_{\alpha \in \mathbb{C}^\times} V_{(n)}^{[\alpha]}$$

for $n \in \mathbb{Z}$, where for $n \in \mathbb{Z}$ and $\alpha \in \mathbb{C}^\times$, if α is an eigenvalue for the operator g on $V_{(n)}$, $V_{(n)}^{[\alpha]}$ is the generalized eigenspace of $V_{(n)}$ for g with the eigenvalue α and if α is not an eigenvalue for g on $V_{(n)}$, $V_{(n)}^{[\alpha]} = 0$. Then

$$V = \coprod_{n \in \mathbb{Z}, \alpha \in \mathbb{C}^\times} V_{(n)}^{[\alpha]}.$$

Let $V^{[\alpha]} = \coprod_{n \in \mathbb{Z}} V_{(n)}^{[\alpha]}$ for $\alpha \in \mathbb{C}^\times$. Then

$$V = \coprod_{\alpha \in \mathbb{C}^\times} V^{[\alpha]}.$$

It is easy to see that the vertex operator algebra V together with this $\mathbb{C}^\times$ is a strongly $\mathbb{C}^\times$-graded vertex operator algebra in the sense of [HLZ2].

For a strongly $\mathbb{C}^\times$-graded vertex operator algebra, a natural module category is the category of strongly $\mathbb{C}^\times$-graded generalized V-modules. Moreover, if the $\mathbb{C}^\times$-grading is given by an automorphism g of V, it is natural to consider strongly $\mathbb{C}^\times$-graded generalized V-modules with an action of g such that the $\mathbb{C}^\times$-gradings are given by generalized eigenspaces for g. The category of the generalizations of g-twisted modules should be a generalization of the category of such strongly $\mathbb{C}^\times$-graded generalized V-modules. In particular, there should also be an action of g and also a $\mathbb{C}^\times$-grading given by generalized eigenspaces for the action of g. Here is our generalization of g-twisted V-module for finite-order g to the general case:

Definition 3.1. Let $(V, Y, \mathbf{1}, \omega)$ be a vertex operator algebra and let g be an automorphism of V. A *strongly $\mathbb{C}^\times$-graded generalized g-twisted V-module* W is a $\mathbb{C} \times \mathbb{C}^\times$-graded vector space $W = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}^\times} W_{[n]}^{[\alpha]}$ equipped with a linear map

$$\begin{aligned} Y^g : V \otimes W &\to W\{x\}[\log x] \\ v \otimes w &\mapsto Y^g(v, x)w \end{aligned}$$

and an action of g satisfying the following conditions for $u, v \in V$ and $w \in W$:

1. The *grading-restriction condition*: For each $n \in \mathbb{C}, \alpha \in \mathbb{C}^\times$, $\dim W_{[n]}^{[\alpha]} < \infty$ and $W_{[n+l]}^{[\alpha]} = 0$ for all sufficiently negative real number l.

2. The *equivariance property*: For $p \in \mathbb{Z}$, $z \in \mathbb{C}^\times$, $v \in V$ and $w \in W$, $Y^{g;p+1}(gv, z)w = Y^{g;p}(v, z)w$, where for $p \in \mathbb{Z}$, $Y^{g;p}$ is the p-th analytic branch of Y^g.

3. The *identity property*: For $w \in W$, $Y^g(\mathbf{1}, x)w = w$.

4. The *duality property*: Let $W' = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}^\times} (W_{[n]}^{[\alpha]})^*$ and, for $n \in \mathbb{C}$, $\pi_n : W \to W_{[n]}$ be the projection. For any $u, v \in V$, $w \in W$ and $w' \in W'$, there exists a multivalued analytic function of the form

$$f(z_1, z_2) = \sum_{i,j,k,l=0}^{N} a_{ijkl} z_1^{m_i} z_2^{n_i} (\log z_1)^k (\log z_2)^l (z_1 - z_2)^{-t}$$

for $N \in \mathbb{N}$, $m_1, \dots, m_N, n_1, \dots, n_N \in \mathbb{C}$ and $t \in \mathbb{Z}_+$, such that the series

$$\langle w', Y^{g;p}(u, z_1)Y^{g;p}(v, z_2)w\rangle = \sum_{n\in\mathbb{C}} \langle w', Y^{g;p}(u, z_1)\pi_n Y^{g;p}(v, z_2)w\rangle, \quad (3.2)$$

$$\langle w', Y^{g;p}(v, z_2)Y^{g;p}(u, z_1)w\rangle = \sum_{n\in\mathbb{C}} \langle w', Y^{g;p}(v, z_2)\pi_n Y^{g;p}(u, z_1)w\rangle, \quad (3.3)$$

$$\langle w', Y^{g;p}(Y(u, z_1 - z_2)v, z_2)w\rangle = \sum_{n\in\mathbb{C}} \langle w', Y^{g;p}(\pi_n Y(u, z_1 - z_2)v, z_2)w\rangle \quad (3.4)$$

are absolutely convergent in the regions $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$, $|z_2| > |z_1 - z_2| > 0$, respectively, to the branch

$$\sum_{i,j,k,l=0}^{N} a_{ijkl} e^{(m_i/k)l_p(z_1)} e^{(n_j/k)l_{p(z_2)}} l_p(z_1)^k l_p(z_2)^l (z_1 - z_2)^{-t}$$

of $f(z_1, z_2)$.

5. The $L^g(0)$- *and* g-*grading conditions*: Let $L^g(n)$ for $n \in \mathbb{Z}$ be the operator on W given by $Y^g(\omega, x) = \sum_{n\in\mathbb{Z}} L^g(n)^{-n-2}$. Then for $w \in W_{[n]}^{[\alpha]}$, there exists $K, \Lambda \in \mathbb{Z}_+$ such that $(L^g(0)-n)^K w = (g-\alpha)^\Lambda w = 0$.

6. The $L(-1)$-*derivative condition*: For $v \in V$,

$$\frac{d}{dx} Y^g(v, x) = Y^g(L(-1)v, x).$$

4 Exponentiating integrals of negative parts of vertex operators

In this section we shall exponentiate integrals of negative parts of vertex operators and prove formulas needed in the next section. We need first several commutator formulas.

The first is the commutator formula for vertex operators which is the special case of the Jacobi identity when we take the coefficients of x_0^{-1} and expand the formal delta-function on the right-hand side explicitly:

Proposition 4.1. *Let $(V, Y, \mathbf{1}, \omega)$ be a vertex operator algebra and (W, Y_W) a weak V-module. The for $u, v \in V$, we have the commutator formula*

$$[Y_W(u, x_1), Y_W(v, x_2)] = \sum_{m\in\mathbb{Z}} \sum_{k\in\mathbb{N}} \binom{m}{k} x_1^{-m-1} x_2^{m-k} Y_W(Y_k(u)v, x_2). \tag{4.1}$$

Let $Y_W^{\leq 2}(u, x)$ be the part of $Y_W(u, x)$ consisting all the monomials in x with powers of x less than or equal to -2, that is,

$$Y_W^{\leq 2}(u, x) = \sum_{m\in\mathbb{Z}_+} Y_m(u) x^{-m-1}.$$

For any vector space W and

$$f(x) = \sum_{n\in\mathbb{Z}\setminus\{-1\}} a_n x^n \in W[[x]] + x^{-2}W[[x^{-1}]],$$

let

$$\int f(x) = \sum_{n\in\mathbb{Z}\setminus\{-1\}} \frac{a_n}{n+1} x^{n+1}.$$

Then we obtain a linear map (*integration*)

$$\int : W[[x]] + x^{-2}W[[x^{-1}]] \to xW[[x]] + x^{-1}W[[x^{-1}]].$$

We have:

Proposition 4.2. *For $u, v \in V$,*

$$\left[\int Y_{\bar{W}}^{\leq 2}(u,x)\Big|_{x=-x_1}, Y_W(v,x_2)\right]$$
$$= Y_W\left(\int Y_{-}^{\leq 2}(u,x)v\Big|_{x=-x_1-x_2}, x_2\right) + \log\left(1+\frac{x_2}{x_1}\right) Y_W(Y_0(u)v, x_2). \tag{4.2}$$

Proof. From (4.1), we obtain

$$\begin{aligned}[Y_{\bar{W}}^{\leq 2}(u,x_1), Y_W(v,x_2)] &= \sum_{m\in\mathbb{Z}_+}\sum_{k\in\mathbb{N}} \binom{m}{k} x_1^{-m-1} x_2^{m-k} Y_W(Y_k(u)v, x_2) \\ &= \sum_{k,l\in\mathbb{N}, k+l\neq 0} \binom{k+l}{k} x_1^{-k-l-1} x_2^{l} Y_W(Y_k(u)v, x_2).\end{aligned} \tag{4.3}$$

Then

$$\left[\int Y_{\bar{W}}^{\leq 2}(u,x)\Big|_{x=-x_1}, Y_W(v,x_2)\right]$$
$$= \sum_{k,l\in\mathbb{N}, k+l\neq 0} \binom{k+l}{k} \frac{(-1)^{k+l+1}}{k+l} x_1^{-k-l} x_2^{l} Y_W(Y_k(u)v, x_2)$$

$$
\begin{aligned}
&= \sum_{k,l\in\mathbb{N},k+l\neq 0} \frac{(k+l)\cdots(k+l-l-1)}{l!}\frac{(-1)^{k+l+1}}{k+l}x_1^{-k-l}x_2^l Y_W(Y_k(u)v,x_2)\\
&= \sum_{k,l\in\mathbb{N},k+l\neq 0} \frac{(-k)\cdots(-k-l+1)}{l!}\frac{(-1)^{k}}{-k}x_1^{-k-l}x_2^l Y_W(Y_k(u)v,x_2)\\
&= \sum_{k\in\mathbb{Z}_+} \frac{(-1)^k}{-k}\sum_{l\in\mathbb{N}}\binom{-k}{l}x_1^{-k-l}x_2^l Y_W(Y_k(u)v,x_2)\\
&\quad + \sum_{l\in\mathbb{Z}_+}\frac{(-1)^{l+1}}{l}x_1^{-l}x_2^l Y_W(Y_0(u)v,x_2)\\
&= \sum_{k\in\mathbb{Z}_+}\frac{(-1)^k}{-k}(x_1+x_2)^{-k}Y_W(Y_k(u)v,x_2)+\log\left(1+\frac{x_2}{x_1}\right)Y_W(Y_0(u)v,x_2)\\
&= Y_W\left(\int Y^{\leq 2}(u,x)v\Big|_{x=-x_1-x_2},x_2\right)+\log\left(1+\frac{x_2}{x_1}\right)Y_W(Y_0(u)v,x_2).
\end{aligned}
$$

∎

We also have:

Proposition 4.3. *For $u\in V$,*

$$
\left[L(-1),\int Y_{\bar{W}}^{\leq 2}(u,y)\Big|_{y=-x}\right] = -\frac{d}{dx}\int Y_{\bar{W}}^{\leq 2}(u,y)\Big|_{y=-x} - Y_0(u)x^{-1}. \tag{4.4}
$$

Proof. By the $L(-1)$-derivative property, we have

$$
[L(-1),Y_W(u,y)] = \frac{d}{dy}Y_W(u,x),
$$

which gives

$$
[L(-1),Y_{\bar{W}}^{\leq 2}(u,y)] = \frac{d}{dy}Y_{\bar{W}}^{\leq 2}(u,y) - Y_0(u)y^{-2}. \tag{4.5}
$$

Note that the integration $\int : W[[x]]+x^{-2}W[[x^{-1}]]\to xW[[x]]+x^{-1}W[[x^{-1}]]$ defined above is invertible and its inverse is in fact the restriction of $\frac{d}{dx}$ to the space $xW[[x]]+x^{-1}W[[x^{-1}]]$. Integrating both sides of (4.5) with respect

to y and then substituting $-x$ for y, we obtain

$$\begin{aligned}\left[L(-1), \int Y_{\bar{W}}^{\leq 2}(u,y)\Big|_{y=-x}\right] &= Y_{\bar{W}}^{\leq 2}(u,y)|_{y=-x} - Y_0(u)x^{-1} \\ &= -\frac{d}{dx}\int Y_{\bar{W}}^{\leq 2}(u,y)\Big|_{y=-x} - Y_0(u)x^{-1}.\end{aligned}$$

∎

Let

$$\widetilde{\Delta}_W^{(u)}(x) = \exp\left(\left(\int Y_{\bar{W}}^{\leq 2}(u,y)\right)\Big|_{y=-x} + Y_0(u)\log x\right).$$

Then we see that $\Delta_W^{(u)}(x) \in (\mathrm{End})[[x^{-1}]][[\log x]]$.

Proposition 4.4. *For $u, v \in V$,*

$$\widetilde{\Delta}_W^{(u)}(x)Y(v,x_2) = Y(v,x_2)(\Delta_W^{(u)}(x+x_2)v, x_2)\Delta_W^{(u)}(x), \tag{4.6}$$

$$[L(-1), \widetilde{\Delta}_W^{(u)}(x)] = -\frac{d}{dx}\Delta_W^{(u)}(x). \tag{4.7}$$

Proof. By (4.2) and (4.4), we obtain (4.6) and (4.7), respectively. ∎

5 Twisted V-modules associated to weight 1 elements of V

In this section, we construct strongly $\mathbb{C}^\times$-graded generalized twisted V-modules associated to $u \in V_{(1)}$ under some reasonable assumptions on V and u.

Given a strongly $\mathbb{C}^\times$-graded generalized V-module, we would like to construct a strongly $\mathbb{C}^\times$-graded generalized twisted V-module by modifying the vertex operator map for the module using the operators $\Delta_V^{(u)}(x)$ constructed in the preceding section. In general $\Delta_V^{(u)}(x)v \in V[[x^{-1}]][[\log x]]$ for $v \in V$. But in our dfinition of strongly $\mathbb{C}^\times$-graded generalized twisted module, the image of the vertex operator map must be in $V\{x\}[\log x]$. So we first need to construct a series $\Delta_V^{(u)}(x)$ and discuss under what assumptions, $\Delta_V^{(u)}(x)v$ is in $V\{x\}[\log x]$ for $v \in V$.

We need the following assumption on the vertex operator algebra V and a fixed element $u \in V_{(1)}$:

Assumption 5.1. We assume that $V_{(0)} = \mathbb{C}\mathbf{1}$ and $V_{(n)} = 0$ for $n < 0$. We assume that $L(1)u = 0$ and there exists an positive integer N_u such that for any $n \in \mathbb{N}$, $Y_0(u)$ as an operator on $V_{(n)}$ have only real eigenvalues and satisfies $(Y_0(u)-\lambda)^{N_u} = 0$ for any eigenvalue λ of the operator $Y_0(u)$ restricted to V_n.

Let $(Y_0(u))_{ss}$ be the semisimple part of $Y_0(u)$, that is, for any generalized eigenvector $v \in V$ for $Y_0(u)$ with eigenvalue λ, $(Y_0(u))_{ss}v = \lambda v$. Then $Y_0(u) = (Y_0(u))_{ss}+(Y_0(u)-(Y_0(u))_{ss})$ and $(Y_0(u)-(Y_0(u))_{ss})^{N_u} = 0$. The commutator formula (4.1) for vertex operators gives

$$[Y_0(u), Y(v, x_2)] = Y(Y_0(u)v, x_2) \tag{5.1}$$

for any $v \in V$. The commutator formula (5.1) gives

$$\begin{aligned} [(Y_0(u))_{ss}, Y(v, x_2)] &= Y((Y_0(u))_{ss}v, x_2), & (5.2)\\ [(Y_0(u) - (Y_0(u))_{ss}), Y(v, x_2)] &= Y((Y_0(u) - (Y_0(u))_{ss})v, x_2. & (5.3) \end{aligned}$$

For any generalized eigenvector $v \in V$ for $Y_0(u)$ with eigenvalue λ, we have $e^{tY_0(u)}v = e^{t(Y_0(u))_{ss}}e^{tY_0(u)-t(Y_0(u))_{ss}}v = e^{t\lambda}e^{tY_0(u)-t(Y_0(u))_{ss}}v$. Since V is spanned by such generalized eigenvector $v \in V$ for $Y_0(u)$, we have a well-defined operator $e^{tY_0(u)}$ on V for every $t \in \mathbb{C}$.

Proposition 5.2. *For $t \in \mathbb{C}$, the operator $e^{tY_0(u)}$ is an automorphism of V.*

Proof. Since $Y_0(u)$ preserve the grading of V, $e^{tY_0(u)}$ also preserve the grading of V. From (5.2) and (5.3), we obtain

$$e^{tY_0(u)}Y(v, x) = Y(e^{tY_0(u)}v, x)e^{tY_0(u)}.$$

The first half of the creation property says that $Y(u, x)\mathbf{1} \in V[[x]]$. In particular, $Y_0(u)\mathbf{1} = 0$. So $e^{tY_0(u)}\mathbf{1} = \mathbf{1}$. Since $u \in V_{(1)}$ and $V_{(n)} = 0$ for $n < 0$, $L(n)u = 0$ for $n \geq 2$. By Assumption 5.1, we also have $L(1)u = 0$. So u is a lowest weight vector of weight 1 and $Y(u, x)$ is a primary field. Thus we have

$$[L(-2), Y(u, x)] = x^{-1}\frac{d}{dx}Y(u, x) - z^{-2}Y(u, x).$$

Taking the coefficients of x^{-1} of both sides of this formula, we obtain

$$[L(-2), Y_0(u)] = 0.$$

Thus $Y_0(u)\omega = L(-2)Y_0(u)\mathbf{1} = 0$ which implies $e^{tY_0(u)}\omega = \omega$. ∎

By the skew-symmetry for V, we have

$$Y(u, x)u = e^{xL(-1)}Y(u, -x)u.$$

Taking Res_x of both sides, we have

$$Y_0(u)u = \sum_{m\in\mathbb{N}} \frac{(-1)^{-m-1}}{m!}(L(-1))^m Y_m(u)u.$$

Since wt $Y_m(u)u = 1 - m - 1 + 1 = 1 - m < 0$ and $V_{(n)} = 0$ when $n < 0$, $Y_m(u)u = 0$ for $m > 1$. Since $V_{(0)} = \mathbb{C}\mathbf{1}$, $Y_1(u)u$ is proportional to $\mathbf{1}$ since wt $Y_1(u)u = 1 - 1 - 1 + 1 = 0$. So $(L(-1))^m Y_m(u)u = 0$ for $m \geq 0$. Thus we obtain $Y_0(u)u = -Y_0(u)u$ which implies $Y_0(u)u = 0$. By the component form of the commutator formula (5.1), we obtain

$$\begin{aligned}[Y_0(u), Y_n(u)] &= Y_n((Y_0(u)u)) \\ &= 0\end{aligned}$$

for $n \in \mathbb{Z}$. Then we can define

$$\begin{aligned}\Delta_V^{(u)}(x) &= x^{Y_0(u)} e^{\int Y^{\leq 2}(u,y)|_{y=-x}} \\ &= x^{(Y_0(u))_{ss}} e^{(Y_0(u)-(Y_0(u))_{ss})\log x} e^{\int Y^{\leq 2}(u,y)|_{y=-x}},\end{aligned}$$

where $(Y_0(u))_{ss}$ is the semisimple part of $Y_0(u)$ and, on an eigenvector of $(Y_0(u))_{ss}$ (that is, a genralized eigenvector of $Y_0(u)$), $x^{(Y_0(u))_{ss}}$ is defined to be x^a if the eigenvalue of the eigenvector is a.

Theorem 5.3. *For any $v \in V$, there exist $m_1, \dots, m_l \in \mathbb{R}$ such that*

$$\Delta_V^{(u)}(x)v \in x^{m_1}V[x^{-1}][\log x] + \cdots + x^{m_l}V[x^{-1}][\log x]$$

and the series $\Delta_V^{(u)}(x)$ satisfies

$$\Delta_V^{(u)}(x)Y(v, x_2) = Y(v, x_2)(\Delta_V^{(u)}(x + x_2)v, x_2)\Delta_V^{(u)}(x) \tag{5.4}$$

and

$$[L(-1), \Delta_V^{(u)}(x)] = -\frac{d}{dx}\Delta_V^{(u)}(x). \tag{5.5}$$

Proof. For any $w \in W$,

$$
\begin{aligned}
&e^{\int Y^{\leq 2}(u,y)v|_{y=-x}} w \\
&\quad = \sum_{k\in\mathbb{N}} \frac{1}{k!}\left(\int Y^{\leq 2}(u,y)|_{y=-x}\right)^k w \\
&\quad = \sum_{k\in\mathbb{N}} \frac{1}{k!}\left(\sum_{n\in\mathbb{Z}_+} \frac{(-1)^n}{-n} Y_n(u)x^{-n}\right)^k w \\
&\quad = \sum_{k\in\mathbb{N}} \frac{1}{k!} \sum_{n_1,\dots,n_k\in\mathbb{Z}_+,\ n_1+\cdots+n_k=l} \frac{(-1)^l}{(-n_1)\cdots(-n_k)} Y_{n_1}(u)\cdots Y_{n_k}(u) w x^{-l}.
\end{aligned}
\tag{5.6}
$$

Since

$$
\begin{aligned}
\text{wt } Y_{n_1}(u)\cdots Y_{n_k}(u) &= (1-n_1-1)+\cdots+(1-n_k-1)) \\
&= -n_1-\cdots-n_k \\
&= -l,
\end{aligned}
$$

we see that $Y_{n_1}(u)\cdots Y_{n_k}(u)w = 0$ when l is sufficiently large. Thus the right-hand side of (5.6) is an element of $W[x^{-1}]$. From Assumption 5.1, our first conclusion

$$
\Delta_V^{(u)}(x)w \in x^{m_1}V[x^{-1}][\log x] + \cdots + x^{m_l}V[x^{-1}][\log x]
$$

follows immediately.

From (4.2), we obtain

$$
\begin{aligned}
&e^{\int Y^{\leq 2}(u,y)v|_{y=-x}} Y(v,x_2) e^{-\int Y^{\leq 2}(u,y)v|_{y=-x}} \\
&\quad = Y\left(e^{\int Y^{\leq 2}(u,y)v|_{y=-x-x_2} + Y_0(u)\log\left(1+\frac{x_2}{x_1}\right)} v, x_2\right) \\
&\quad = Y\left(e^{Y_0(u)\log\left(1+\frac{x_2}{x_1}\right)} e^{\int Y^{\leq 2}(u,y)v|_{y=-x-x_2}} v, x_2\right).
\end{aligned}
\tag{5.7}
$$

On the other hand, by (5.2) and (5.3), we have

$$
\begin{aligned}
&x^{Y_0(u)} Y(v,x_2) \\
&\quad = x^{(Y_0(u))_{ss}} e^{(Y_0(u)-(Y_0(u))_{ss})\log x} Y(v,x_2) \\
&\quad = Y(x^{(Y_0(u))_{ss}} e^{(Y_0(u)-(Y_0(u))_{ss})\log x} v, x_2) x^{(Y_0(u))_{ss}} e^{(Y_0(u)-(Y_0(u))_{ss})\log x} \\
&\quad = Y(x^{Y_0(u)} v, x_2) x^{Y_0(u)}.
\end{aligned}
\tag{5.8}
$$

Using (5.7) and (5.8), we obtain (5.4).

Finally, notice that both $[L(-1),\cdot]$ and $-\frac{d}{dx}$ are derivatives on the space $\mathbb{C}[[\int Y^{\leq 2}(u,y)|_{y=-x}]]$ of power series in $\int Y^{\leq 2}(u,y)|_{y=-x}$. So from (4.4), we obtain

$$\begin{aligned}
&[L(-1), e^{\int Y^{\leq 2}(u,y)|_{y=-x}}] + \frac{d}{dx} e^{\int Y^{\leq 2}(u,y)|_{y=-x}} \\
&\quad = \left[L(-1), \int Y^{\leq 2}(u,y)|_{y=-x}\right] e^{\int Y^{\leq 2}(u,y)|_{y=-x}} \\
&\qquad + \left(\frac{d}{dx}\int Y^{\leq 2}(u,y)|_{y=-x}\right) e^{\int Y^{\leq 2}(u,y)|_{y=-x}} \\
&\quad = -Y_0(u)x^{-1} e^{\int Y^{\leq 2}(u,y)|_{y=-x}}. \qquad (5.9)
\end{aligned}$$

The equality (5.9) is equivalent to (5.5). ∎

Let $W = \coprod_{n\in\mathbb{C}} W_{\langle n\rangle}$ be a strongly $\mathbb{C}^\times$-graded generalized V-module with an action of $e^{(Y_W)_0(u)}$ such that the $\mathbb{C}^\times$-grading is given by generalized eigenspaces for the action of $e^{(Y_W)_0(u)}$. We define a linear map

$$\begin{aligned}
Y_W^{(u)} : V\otimes W &\to W\{x\}[\log x] \\
v\otimes w &\mapsto Y_W^{(u)}(v,x)w
\end{aligned}$$

by

$$Y_W^{(u)}(v,x)w = Y_W(\Delta_V^{(u)}(x)v, x).$$

First, we have:

Lemma 5.4. *The vertex operator $Y_W^{(u)}(\omega,x)$ is in $W[[x,x^{-1}]]$ and if we write*

$$Y_W^{(u)}(\omega,x) = \sum_{n\in\mathbb{Z}} L_W^{(u)}(n)x^{-n-2},$$

then $L_W^{(u)}(0) = L_W(0) + (Y_W)_0(u) - \frac{1}{2}\mu$ where $\mu\in\mathbb{C}$ is given by $(Y_W)_1(u)u = \mu\mathbf{1}$ (note that $(Y_W)_1(u)u \in V_{(0)}$ and by Assumption 5.1, it must be proportional to $\mathbf{1}$). In particular, for $n\in\mathbb{C}$ and $\alpha\in\mathbb{C}^\times$, $W^{[\alpha]}_{\langle n-\alpha+\frac{1}{2}\mu\rangle}$ is the generalized eigenspace for both $L_W^{(u)}(0)$ and $e^{(Y_W)_0(u)}$ with the eigenvalues n and α, respectively.

Proof. By (4.1), we have

$$[Y(\omega, x_1), Y(u, x_2)] = \mathrm{Res}_{x_0} x_2^{-1} \delta\left(\frac{x_1 - x_0}{x_2}\right) Y(Y(\omega, x_0)u, x_2).$$

Taking the coefficients of x_1^0 in both sides and noticing that from our assumption, $L(n)u = 0$ for $n > 0$, we obtain

$$\begin{aligned}[L(-2), Y(u, x_2)] &= -x_2^{-1} Y(L(-1)u, x_2) + x_2^{-2} Y(L(0)u, x_2) \\ &= -x_2^{-1} \frac{d}{dx_2} Y(u, x_2) + x_2^{-2} Y(u, x_2).\end{aligned}$$

Taking the coefficients of x_2^{-m-1} in both sides, we obtain

$$[L(-2), Y_m(u)] = mY_{m-2}(u)$$

for $m \in \mathbb{Z}$. Then we have

$$\begin{aligned}Y_m(u)\omega &= Y_m(u)L(-2)\mathbf{1} \\ &= L(-2)Y_m(u)\mathbf{1} - mY_{m-2}(u)\mathbf{1}\end{aligned}$$

for $m \in \mathbb{Z}$. Since $Y_m(u)\mathbf{1} = 0$ for $m \geq 0$ and $Y_{-1}(u)\mathbf{1} = u$, we obtain

$$Y_m(u)\omega = 0$$

for $m \in \mathbb{N}$ but $m \neq 1$ and

$$Y_1(u)\omega = -u.$$

In particular, $Y_0(u)\omega = (Y_0(u))_{ss}\omega$. We also know that $Y_m(u)u = 0$ for $m > 1$ and $Y_1(u)u = \mu\mathbf{1}$. So

$$e^{\int Y^{\leq 2}(u,y)|_{y=-x}}\omega = \omega + ux^{-1} - \frac{1}{2}\mu\mathbf{1}x^{-2}$$

and

$$\begin{aligned}\Delta_V^{(u)}(x)\omega &= x^{(Y_0(u))_{ss}}(\omega + ux^{-1} - \frac{1}{2}\mu\mathbf{1}x^{-2}) \\ &= \omega + ux^{-1} - \frac{1}{2}\mu\mathbf{1}x^{-2}.\end{aligned}$$

Thus

$$Y_W^{(u)}(\omega, x) = Y_W(\omega, x) + Y(u, x)x^{-1} - \frac{1}{2}\mu x^{-2} \in W[[x, x^{-1}]]$$

and $L_W^{(u)}(0) = L_W(0) + (Y_W)_0(u) - \frac{1}{2}\mu$.

The second conclusion follows immediately. ∎

We have the following consequence:

Corollary 5.5. *The space W is also doubly graded by the generalized eigenvalues of $L_W^u(0)$ and $e^{(Y_W)_0(u)}$, that is, W has a $\mathbb{C}\times\mathbb{C}^\times$-grading $W = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}^\times} W_{[n]}^{[\alpha]}$ where*

$$W_{[n]}^{[\alpha]} = W_{\langle n-\alpha+\frac{1}{2}\mu\rangle}^{[\alpha]}$$

such that $w \in W_{[n]}^{[\alpha]}$ for $n \in \mathbb{C}$ and $\alpha \in \mathbb{C}^\times$ is a generalized eigenvector for $L_W^u(0)$ with the eigenvalue n and a generalized eigenvector for $e^{(Y_W)_0(u)}$ with the eigenvalue α.

Proof. By Lemma 5.4, every element of W is a finite sum of vectors which are generalized eigenvecotrs for both $L_W^{(u)}(0)$ and $(Y_W)_0(u)$, or equivalently, $e^{(Y_W)_0(u)}$. ∎

Theorem 5.6. *The pair $(W, Y_W^{(u)})$, where W is equipped with the $\mathbb{C}\times\mathbb{C}$-grading given in Corollary 5.5, is a strongly $\mathbb{C}^\times$-graded generalized $e^{2\pi i Y_0(u)}$-twisted V-module.*

Proof. We first prove that W satisfies the grading-restriction condition. By Lemma 5.4, $L_W(0) = L_W^{(u)}(0) - (Y_W)_0(u) + \frac{1}{2}\mu$. Since the generalized V-module W satisfies the grading-restriction condition, $\dim W_{\langle n\rangle}^{[\alpha]} < \infty$ for $n \in \mathbb{C}$ and $\alpha \in \mathbb{C}^\times$ and for any fixed $n \in \mathbb{C}$ and $\alpha \in \mathbb{C}^\times$, $W_{\langle n+l\rangle} = 0$ when l is a sufficiently negative integer. So for $n \in \mathbb{C}$, $\alpha \in \mathbb{C}^\times$, $W_{[n]}^{[\alpha]} = W_{\langle n-\alpha+\frac{1}{2}\mu\rangle}^{[\alpha]}$ are finite dimensional and for any fixed $n, \alpha \in \mathbb{C}$ $W_{[n+l]}^{[\alpha]} = W_{\langle n+l-\alpha+\frac{1}{2}\mu\rangle} = 0$ when l is a sufficiently negative integer.

From the first part of the creation property, $Y(u,x)\mathbf{1} \in V[[x]]$ for vertex operator algebra, we have $Y_n(u)\mathbf{1} = 0$ for $n \geq 0$. In particular, $(Y_0(u))_{ss}\mathbf{1} = (Y_0(u) - (Y_0(u))_{ss})\mathbf{1} = 0$. We obtain

$$\begin{aligned}\Delta_V^{(u)}(x)\mathbf{1} &= x^{(Y_0(u))_{ss}} e^{(Y_0(u)-(Y_0(u))_{ss})\log x} e^{\int Y^{\leq 2}(u,y)|_{y=-x}}\mathbf{1}\\ &= \mathbf{1}.\end{aligned}$$

For $w \in W$,

$$\begin{aligned} Y_W^{(u)}(\mathbf{1}, x)w &= Y_W(\Delta_V^{(u)}(x)\mathbf{1}, x)w \\ &= Y_W(\mathbf{1}, x)w \\ &= w, \end{aligned}$$

proving the identity property.

Let $v \in V$ be a generalized eigenvector for $Y_0(u)$ with eigenvalue λ. Then for $v \in V$ and $z \in \mathbb{C}^\times$,

$$\begin{aligned} &\Delta_V^{(u)}(x)e^{2\pi i Y_0(u)}v\Big|_{x^n=e^{nl_p(z)},\ \log x=l_p(z)} \\ &\quad = x^\lambda e^{(Y_0(u)-(Y_0(u))_{ss})\log x}e^{2\pi i\lambda}e^{2\pi i(Y_0(u)-(Y_0(u))_{ss})}v\Big|_{x^n=e^{nl_p(z)},\ \log x=l_p(z)} \\ &\quad = x^\lambda e^{(Y_0(u)-(Y_0(u))_{ss})\log x}v\Big|_{x^n=e^{nl_{p+1}(z)},\ \log x=l_{p+1}(z)} \\ &\quad = \Delta_V^{(u)}(x)v\Big|_{x^n=e^{nl_{p+1}(z)},\ \log x=l_{p+1}(z)}. \end{aligned}$$

Thus

$$\begin{aligned} &(Y_W^{(u)})^{e^{2\pi i Y_0(u)};p}(e^{2\pi i Y_0(u)}v, z)w \\ &\quad = Y_W(\Delta_V^{(u)}(x)e^{2\pi i Y_0(u)}v, x)w\Big|_{x^n=e^{nl_p(z)},\ \log x=l_p(z)} \\ &\quad = Y_W(\Delta_V^{(u)}(x)v, x)w\Big|_{x^n=e^{nl_{p+1}(z)},\ \log x=l_{p+1}(z)} \\ &\quad = (Y_W^{(u)})^{e^{2\pi i Y_0(u)};p+1}(v, z)w \end{aligned}$$

for $v \in V$ and $w \in W$, that is, the equivariance property holds.

By Corollary 5.5, the $L^{(u)}(0)$- and $e^{2\pi i Y_0(u)}$-grading condition is satisfied.

By the $L(-1)$-derivative property for the vertex operator map Y_W and (5.5), we have

$$\begin{aligned} \frac{d}{dx}Y_W^{(u)}(v, x) &= \frac{d}{dx}Y_W(\Delta_V^{(u)}(x)v, x) \\ &= Y_W(L(-1)\Delta_V^{(u)}(x)v, x) + Y_W\left(\left(\frac{d}{dx}\Delta_V^{(u)}(x)v\right), x\right) \\ &= Y_W(L(-1)\Delta_V^{(u)}(x)v, x) + Y_W(-[L(-1), \Delta_V^{(u)}(x)]v), x) \\ &= Y_W(\Delta_V^{(u)}(x)L(-1)v, x) \\ &= Y_W^{(u)}(L(-1)v, x), \end{aligned}$$

for $v \in V$, proving the $L(-1)$-derivative property.

Finally we prove the duality property. By Corollary 5.5, $W_{[n]}^{[\alpha]} = W_{\langle n-\alpha+\frac{1}{2}\mu\rangle}^{[\alpha]}$. So the graded dual of W with respect to the grading $W = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}^\times} W_{\langle n\rangle}^{[\alpha]}$ and the graded dual with respect the new $\mathbb{C}\times\mathbb{C}^\times$-grading $W = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}^\times} W_{[n]}^{[\alpha]}$ in Corollary 5.5 are equal as vector spaces, though their gradings are different. We shall use W' to denote the underlying vecotr space of these two graded dual. It will be clear from the context which grading we will be using.

For $v_1, v_2 \in V$, $w \in W$ and $w' \in W'$, we know that there exist $m_1, \dots, m_r$, $n_1, \dots, n_s \in \mathbb{R}$ such that

$$\Delta_V^{(u)}(x_1)e^{2\pi i Y_0(u)}v_1 \in x_1^{m_1}V[x_1^{-1}][\log x_1] + \cdots + x_1^{m_r}V[x_1^{-1}][\log x_1]$$

and

$$\Delta_V^{(u)}(x_2)e^{2\pi i Y_0(u)}v_2 \in x_2^{n_1}V[x_2^{-1}][\log x_2] + \cdots + x_2^{n_s}V[x_2^{-1}][\log x_2].$$

Thus, using the rationality, commutativity and associativity properties for the V-module W and the fact that in the region $|z_2| > |z_1 - z_2| > 0$

$$\begin{aligned}&\Delta_V^{(u)}(x_2+x_0)v_1\Big|_{x_0^n=(z_1-z_2)^n,\ x_2^n=e^{nl_p(z_2)},\ \log x_2=l_p(z_2)}\\ &\quad= \Delta_V^{(u)}(x_1)v_1\Big|_{x_1^n=e^{nl_p(z_1)},\ \log x_1=l_p(z_1)},\end{aligned}$$

we see that there exists $a_{ijkl} \in \mathbb{C}$ for $i,j,k,l = 0,\dots,N$, $\alpha_i, \beta_j \in \mathbb{C}$ for $i,j = 0,\dots,N$ and $t \in \mathbb{N}$ such that the series

$$\begin{aligned}&\langle w', (Y_W^{(u)})^{e^{2\pi i Y_0(u)};p}(v_1,z_1)(Y_W^{(u)})^{e^{2\pi i Y_0(u)};p}(v_2,z_2)w\rangle\\ &\quad= \langle w', Y_W(\Delta_V^{(u)}(x_1)v_1, x_1)\cdot\\ &\qquad\qquad \cdot Y_W(\Delta_V^{(u)}(x_2)v_2,x_2)w\rangle\Big|_{x_1^n=e^{nl_p(z_1)},\ \log x_1=l_p(z_1),\ x_2^n=e^{nl_p(z_2)},\ \log x_2=l_p(z_2)},\\ &\langle w', (Y_W^{(u)})^{e^{2\pi i Y_0(u)};p}(v_2,z_2)(Y_W^{(u)})^{e^{2\pi i Y_0(u)};p}(v_1,z_1)w\rangle\\ &\quad= \langle w', Y_W(\Delta_V^{(u)}(x_2)v_2, x_2)\cdot\\ &\qquad\qquad \cdot Y_W(\Delta_V^{(u)}(x_1)v_1,x_1)w\rangle\Big|_{x_1^n=e^{nl_p(z_1)},\ \log x_1=l_p(z_1),\ x_2^n=e^{nl_p(z_2)},\ \log x_2=l_p(z_2)},\\ &\quad\langle w', Y_W(Y(\Delta_V^{(u)}(x_2+x_0)v_1, x_0)\cdot\\ &\qquad\qquad \cdot\Delta_V^{(u)}(x_2)v_2,x_2)w\rangle\Big|_{x_0^n=(z_1-z_2)^n,\ x_2^n=e^{nl_p(z_2)},\ \log x_2=l_p(z_2)}\end{aligned}$$

are absoluely convergent in the regions $|z_1| > |z_2| > 0$, $|z_1| > |z_2| > 0$, $|z_2| > |z_1 - z_2| > 0$, respectively, to the branch

$$\sum_{i,j,k,l=0}^{N} a_{ijkl} e^{m_i l_p(z_1)} e^{n_j l_{p(z_2)}} l_p(z_1)^k l_p(z_2)^l (z_1 - z_2)^{-t}$$

of the multivalued analytic function

$$\sum_{i,j,k,l=0}^{N} a_{ijkl} z_1^{m_i} z_2^{n_j} (\log z_1)^k (\log z_2)^l (z_1 - z_2)^{-t}.$$

By (5.4),

$$\begin{aligned}
&\langle w', Y_W(Y(\Delta_V^{(u)}(x_2 + x_0)v_1, x_0)\cdot \\
&\qquad\qquad \cdot \Delta_V^{(u)}(x_2)v_2, x_2)w\rangle\Big|_{x_0^n=(z_1-z_2)^n,\ x_2^n=e^{nl_p(z_2)},\ \log x_2=l_p(z_2)} \\
&= \langle w', Y_W(\Delta_V^{(u)}(x_2)Y(v_1, x_0)v_2, x_2)w\rangle\Big|_{x_0^n=(z_1-z_2)^n,\ x_2^n=e^{nl_p(z_2)},\ \log x_2=l_p(z_2)} \\
&= \langle w', (Y_W^{(u)})^{e^{2\pi i Y_0(u)};p}(Y(v_1, z_1 - z_2)v_2, z_2)w\rangle.
\end{aligned}$$

Thus the suality property is proved. ∎

We have the following immediate consequence:

Corollary 5.7. *The correspondence given by $(W, Y_W) \to (W, Y_W^{(u)})$ is a functor from the category of strongly $\mathbb{C}^\times$-graded generalized V-modules to the category of strongly $\mathbb{C}^\times$-graded generalized $e^{2\pi i Y_0(u)}$-twisted V-modules.* ∎

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